VALAY SAITWADEKAR

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EDUCATION

California State University Fullerton, United States of America **Expected: May 2026**
Candidate for Masters of Science in Computer Science CGPA 3.96/4

Shree L. R. Tiwari College of Engineering, Mumbai University, India **May 2024**
Bachelor of Engineering in Computer Engineering with CGPA 9.11/10
Honors in Cyber Security with *Distinction* (3.65/4)

PROFESSIONAL EXPERIENCE

Filtroll International (India) Pvt. Ltd., Mumbai, India **December 2022 – June 2023**
Intern – Software Engineer

- Implemented inventory management, accounting, and CRM modules using Python and Flask within the ERP system to help the business manage inventory levels and track financial transactions more efficiently, along with custom reports, dashboards, and analytics to provide business insights, reducing manual work by 50% and improving inventory management accuracy by 85%.
- Formed ERP system requirements and specifications document based on meetings with key business stakeholders to understand their needs and planned development of new capabilities aligned with business growth.
- Designed a relational database schema in MySQL to organize business data for reports and analysis with stored procedures and database triggers to enforce business logic and data integrity.
- Created user guides and admin manuals to support onboarding new users and documentation for users covering system navigation and module functions and conducted training to help employees adopt the new ERP tools.

PROJECTS

DupliCron – Metadata-Based Incremental File Synchronization and Backup

- Built a C++20 tool for Windows and Linux that performs incremental backups from multiple source directories by detecting file changes using file size and modification time, with a binary metadata cache to prevent redundant copying across runs.
- Parallelized scanning, hashing, and copying phases using a custom thread pool to efficiently manage concurrent tasks, maximizing resource usage and reducing synchronization time through overlapping operations. Tested with over a million files, maintaining a low memory footprint of just a few megabytes.
- Designed a disk-aware per-source copy mechanism using thread-safe queues with serialized and parallel modes for peak throughput. Ensured data integrity by updating metadata only after verified copies and enabling automatic retries on failure.
- Implemented structured logging, UTF-8 support for multilingual file and folder names and configured the system to be fully driven by a user-editable file supporting flexible sync modes, source exclusions, and stale file cleanup.

TransactViaMe – Event-Driven Payment Backend Platform

- Built and orchestrated asynchronous payment workflows across authorization, settlement and ledger services using Java and Python, connected via Kafka, PostgreSQL and Redis for durable state and caching, exposing functionality through REST APIs.
- Implemented high-throughput Kafka pipelines processing 50K+ payment events per minute, with idempotent consumers, retries, and failure recovery, ensuring data integrity during restarts and load spikes.
- Enabled automatic horizontal scaling on AWS EKS using Kubernetes with Horizontal Pod Autoscaler, maintaining API response times under 200 ms while handling synthetic stress testing with k6/Locust.
- Developed supporting services and tooling in TypeScript, and validated system reliability through stress and failure testing, including pod restarts and Kafka consumer rebalancing, with observability via Prometheus, Grafana, and centralized logging.

DISHA – Deep Learning based Cyclone Intensity Estimation with NLP based Report Generation

- Launched a website that integrates a CNN model to classify cyclone intensity with 94.5% accuracy using IR(Infrared) images from the MOSDAC website, categorizing cyclones into levels 1 to 5, and providing real-time insights into cyclone severity.
- Trained a T5 transformer model to generate multilingual safety guidelines and precautionary measures tailored to each cyclone level, enhancing accessibility and improving response strategies for diverse audiences.
- Setup Twilio API sending SMS alerts with cyclone intensity and precautionary guidelines, providing timely information to users and OpenWeatherMap API providing additional weather-related information.

TECHNICAL SKILLS

- Programming Languages: C, C++, Java, Kotlin, Python, Shell Scripting, R, HTML, CSS, JavaScript, TypeScript
- Tools: Visual Studio, Android Studio, WSL, Cmake, gdb, perf, gprof, valgrind, KCachegrind, Google Benchmark, Wireshark, Nmap, Burp Suite, OWASP ZAP, Postman, Docker, Kubernetes, Git, Terraform, Ansible, Jenkins
- Frameworks & Platforms: AWS, Redis, Kafka, Prometheus, Grafana, ELK Stack, Maven, Gradle, FastAPI, NestJS, Flask, Django, Spring Boot, jQuery, Node.js, PostgreSQL, MySQL

RELEVANT COURSEWORK

- Operating Systems Design
- Advanced Computer Networking
- Software Design & Architecture
- Software Maintenance
- AI, ML, NLP and Social Media Analysis
- Digital Logic & Comp Arch
- Distributed Computing
- Microprocessors
- Security Information Management
- Application & System Security
- Cyber Security and Law
- Web Security